Prompt 9:

Objective: Detect the incorporation of specific sustainable practices within the provided source code, focusing on green architectural tactics for machine learning. Analyze the code to identify direct and indirect sustainable approaches.

Task Description: 1) Review the source code to determine whether each of the following sustainable approaches mentioned or implemented:

1.Choose an energy efficient algorithm: Choose the most energy-efficient algorithm that achieves sufficient level of accuracy

2.Choose a lightweight algorithm alternative: If possible, choose lighter alternatives of existing algorithms

3.Decrease model complexity: Decrease the complexity of an ML model

4.Consider reinforcement learning for energy efficiency: Use reinforcement learning to optimize energy efficiency at run time

5.Use dynamic parameter adaptation: Design parameters that are dynamically adapted based on the input data

6.Use built-in library functions: Use built-in libraries for ML models if possible

7.Set energy consumption as a model constraint: Consider energy consumption as one predetermined parameter for optimizing the ML model

8.Consider graph substitution: Replace energy-intensive model parts with similar, but less energy-consuming parts

9.Enhance model sparsity: Reduce the number of model parameters or set their values to zero

10.Consider energy aware pruning: Prune neural networks starting from the most energy intensive layer

11.Consider transfer learning: Use pre-trained ML models for other similar tasks

12.Consider knowledge distillation: Use knowledge from a large ML model to train a smaller model

13.Use quantization-aware training: Convert high-precision data types to lower precision during training

14.Use checkpoints during training: Use checkpoints to avoid a knowledge loss in case of a premature termination

15.Design for memory constraints: Consider possible memory constraints during training

2) Additionally, identify any new direct or indirect sustainable practices not listed above that are present in the code.

Additional Instructions: Focus on specific lines or sections that relate to the listed sustainable practices. Clearly state whether each approach is present, supported by specific code snippets. If new approaches are not present, only write "No" without describing the code. Reporting Instructions: Structure the output as follows: Presence of Approaches: For each approach found in the source code, specify its occurrence and provide the relevant code snippet. If not present, write “No”. New Approaches: Analyze the code to identify new direct and indirect sustainable practices. If a new approach is found, write "Yes" including the code and a brief description. If not present, write “No”.

Examples: "Design for memory constraints: No" "Use built-in library function: import torch" "New Approaches: No" "New Approaches: Yes, Implementing Early Stopping early_stopping = EarlyStopping(monitor='val_loss')"

Prompt 9 riadattato:

Objective: Rilevare l’utilizzo di specifiche pratiche sostenibili nel codice sorgente fornito, concentrandosi su tattiche architetturali green per sistemi AI-Enabled. Analizzare il codice per identificare approcci sostenibili diretti e indiretti.

Task Description: Esaminare il codice sorgente per determinare se ciascuno dei seguenti approcci sostenibili è menzionato o implementato:

1. Utilizzare funzioni di libreria integrate: utilizzare librerie integrate per i modelli ML, se possibile.

2. Considerare la energy aware pruning: potare le reti neurali partendo dallo strato più energivoro.

3. Utilizzare i checkpoint durante la formazione: utilizzare i checkpoint per evitare una perdita di conoscenze in caso di interruzione prematura.

Additional Instruction: Basare il giudizio esclusivamente su ciò che è osservabile nello snippet, senza assumere dettagli non presenti; concentrarsi su righe o sezioni specifiche relative alle pratiche sostenibili elencate. Indicare chiaramente se ciascun approccio è presente, supportato da frammenti di codice specifici. Se non sono presenti nuovi approcci, scrivere semplicemente "No" senza descrivere il codice.

Reporting Instruction: Strutturare l'output come segue: Presenza di approcci: per ogni approccio trovato nel codice sorgente, specificarne l'occorrenza e fornire il frammento di codice pertinente. Se non presente, scrivere "No".

Examples: Esempio di esito per uno snippet sottoposto all'llm: "Built-in library: assente"; "Pruning: assente"; "Training Checkpoint: presente, torch.save(net.state_dict(), "ckpt_best_val.pt")".